SHWETHA V

Undergraduate Researcher, Mechanical Engineering ⋄ Indian Institute of Technology Madras, Chennai, India
shwetha.vishy@gmail.com ⋄ Website ⋄ GitHub

RESEARCH INTERESTS

Computational Fluid Dynamics (CFD)
Thermoacoustic Instability and Combustion Dynamics
Nonlinear Dynamics and Complex Systems
High Performance Scientific Computing

EDUCATION

Indian Institute of Technology Madras (IITM) — 2023 - 2027
B.Tech. in Mechanical Engineering — CGPA: 8.2/10
Minor: Physics

Vanavani Matriculation Higher Secondary School, Chennai — 2023
Class XII: 99%

INTERNATIONAL ACADEMIC EXPOSURE

Computational Physics Spring School — 2026
Okinawa Institute of Science and Technology (Online)

Sakura Science Exchange Program — 2025
University of Tokyo, Japan

MEDIA RECOGNITION

Interviewed by **NHK (Japan Broadcasting Corporation)** as the top student of my school and featured in a published news article. Print. Video.

RESEARCH EXPERIENCE

Thermoacoustic Instability Analysis - CTCS, IIT Madras — Jan 2025 - Present
Supervisor: **Prof. R. I. Sujith**

- Investigating abrupt transitions to thermoacoustic instability in turbulent combustors using nonlinear time-series analysis and complex network methods to study fluid–acoustic coupling.
  First-author manuscript in preparation.
- Analyzing experimental datasets obtained from high-speed chemiluminescence imaging and particle image velocimetry (PIV).
- Co-author: "Periodic modulation of the space-filling nature of the turbulent flame leads to spiky heat release oscillations." arXiv preprint (2025) https://doi.org/10.48550/arXiv.2511.15770.

ENGINEERING PROJECTS

Autonomous Robot Path Planning - Team Abhiyaan (CFI IITM) — Mar 2024 - Aug 2024

- Developed Voronoi-based path planning algorithm for autonomous hexagonal robots.
- Implemented navigation algorithm in C++ and integrated it with the RoboCup simulation environment.
- Conducted testing in collaboration with the mechatronics subsystem team.

Mechanical Engineering Course Projects

- Designed rear-wheel transmission system for a Mahindra XUV automobile.
- Developed a MATLAB/Simulink controller and observer for a two-dimensional self-balancing robot.
- Built an Arduino-based autonomous fire-fighting robot.
- Developed a machine-learning framework for chatter detection in boring-bar machining processes.

3D Printed Resizable Musical Instruments (Veena) - IIT Madras Dec 2025 - Present

- Developed parametric CAD models using SolidWorks.
- Studying fret placement and octave scaling through mathematical modelling.

RELEVANT COURSEWORK

Advanced Computational Fluid Dynamics (CFD solver development in Fortran)
Parallel Scientific Computing (MPI, OpenMP, Linux HPC workflows)
Acoustic Instabilities in Aerospace Propulsion
Measurements, Instrumentation and Control Systems
Linear Algebra, Classical Mechanics, Quantum Physics

TECHNICAL SKILLS

Programming: C, C++, Fortran, Python, MATLAB, Octave

Scientific Computing: Linux, Bash, MPI, OpenMP, ParaView

Engineering Software: SolidWorks, Creo, AutoCAD, Simulink

Other Tools: LaTeX, HTML, CSS, SQL, Node.js, Figma

LANGUAGES

English – Fluent.
Japanese – Basic conversational (IIT Madras Japanese course – top student).

SCHOLASTIC ACHIEVEMENTS

- All India Rank 3472 - JEE Advanced.
- All India Rank 2251 - JEE Main (among $\sim$1 million candidates).
- International Rank 147 - NSTSE Olympiad.
- All India Rank 46 - Technothlon (IIT Guwahati).
- Medalist in SOF Olympiads (IEO, NSO, IMO) throughout school.

WORK EXPERIENCE

Software Development Intern - MachIntell Feb 2024 - Jun 2024

- Developed APIs for reliability testing modules used by industrial equipment manufacturers.
- Collaborated with engineers on backend integration and product development.

LEADERSHIP AND VOLUNTEERING

Volunteer - Saṁskṛta Samiti, IIT Madras 2025 - Present

Conduct Sanskrit learning sessions and assist in educational outreach activities.

Associate Manager - Entrepreneurship Cell (E-Cell), IIT Madras

Organized entrepreneurship competitions and innovation events.

President - Interact Club (Rotary) 2021 - 2022

Led student initiatives including personal development programs, career counseling and community service drives.

EXTRACURRICULAR ACTIVITIES

- Carnatic vocalist with 12+ years of training.
- Completed a full marathon (42.195 km) in 2026.
- Poetry published in literary anthologies.